

BCA MATHEMATICS-I (BCA-0101)

Complete Unit-I Study Material

As per Himachal Pradesh University (HPU) BCA Semester-I Syllabus

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1 Set Theory

1.1 Definition of Set

A **set** is a well-defined collection of distinct objects.

Examples:

- $A = \{1, 2, 3, 4, 5\} \rightarrow$ Set of first 5 natural numbers
- $B = \{a, e, i, o, u\} \rightarrow$ Set of vowels

1.2 Types of Sets

Type	Definition	Example
Finite Set	Contains countable elements	$A = \{2, 4, 6, 8\}$
Infinite Set	Contains uncountable elements	$N = \{1, 2, 3, \dots\}$
Empty Set ($\emptyset$)	Contains no elements	$\{x \mid x^2 = 4, x \text{ is odd}\}$
Universal Set (U)	Contains all elements under consideration	$U = \{1, 2, 3, \dots, 10\}$

1.3 Representation of Sets

Roster Form: Listing all elements

$$A = \{1, 2, 3, 4, 5\}$$

Set Builder Form: Describing property

$$A = \{x \mid x \in N, x \leq 5\}$$

1.4 Set Operations

Operation	Symbol	Meaning	Example ($A=\{1,2,3,4\}$, $B=\{3,4,5,6\}$)
Union	$A \cup B$	Elements in A OR B	$\{1,2,3,4,5,6\}$
Intersection	$A \cap B$	Elements in A AND B	$\{3,4\}$
Difference	$A - B$	Elements in A but NOT in B	$\{1,2\}$
Complement	A'	Elements NOT in A (within U)	$\{5,6,7,8\}$

1.5 Power Set

The set of all subsets of a given set.

Formula: If $|A| = n$, then $|P(A)| = 2^n$

Example: For $A = \{a, b, c\}$

- $P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$
- Number of elements $= 2^3 = 8$

1.6 Important Laws

De Morgan's Laws:

- $(A \cup B)' = A' \cap B'$
- $(A \cap B)' = A' \cup B'$

Distributive Laws:

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

2 Relations

2.1 Cartesian Product

Definition: $A \times B = \{(a, b) \mid a \in A, b \in B\}$

Formula: $|A \times B| = |A| \times |B|$

Example: $A = \{1, 2\}$, $B = \{a, b, c\}$

- $A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$
- $|A \times B| = 2 \times 3 = 6$

2.2 Definition of Relation

A relation R from set A to set B is a subset of $A \times B$.

- If $R \subseteq A \times B$, then R is a relation from A to B .
- If $R \subseteq A \times A$, then R is a relation on A .

2.3 Properties of Relations

Property	Definition	Mathematical Condition
Reflexive	Every element relates to itself	$(a, a) \in R$ for all $a \in A$
Symmetric	If aRb , then bRa	$(a, b) \in R \rightarrow (b, a) \in R$
Transitive	If aRb and bRc , then aRc	$(a,b), (b,c) \in R \rightarrow (a,c) \in R$

2.4 Equivalence Relation

A relation that is **Reflexive + Symmetric + Transitive**.

Example: $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$ on $A = \{1,2,3\}$

2.5 Transitive Closure

The smallest transitive relation containing R .

Example: $R = \{(1,2), (2,3), (3,4)\}$

- Transitive Closure = $\{(1,2), (2,3), (3,4), (1,3), (2,4), (1,4)\}$

3 Quadratic Equations

3.1 Standard Form

$$ax^2 + bx + c = 0, \text{ where } a \neq 0$$

3.2 Methods of Solving

Method 1: Factorization

- Split the middle term
- Example: $x^2 - 5x + 6 = 0 = (x - 2)(x - 3) = 0 \therefore x = 2, 3$

Method 2: Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Method 3: Completing the Square

- Convert $ax^2 + bx + c = 0$ to $(x + p)^2 = q$

3.3 Discriminant

$$D = b^2 - 4ac$$

Condition	Nature of Roots
$D > 0$	Two distinct real roots
$D = 0$	One real root (equal roots)
$D < 0$	No real roots (complex roots)

3.4 Sum and Product of Roots

If α and β are roots of $ax^2 + bx + c = 0$:

- **Sum** $(\alpha + \beta) = -b/a$
- **Product** $(\alpha\beta) = c/a$

To form quadratic equation: $x^2 - (\text{sum})x + \text{product} = 0$

4 Sequence & Series

4.1 Arithmetic Progression (AP)

Definition: Constant difference between consecutive terms.

- First term = a
- Common difference = d

Important Formulas:

- **nth term:** $a_n = a + (n - 1)d$
- **Sum of n terms:** $S_n = n/2[2a + (n - 1)d]$
- **Sum of n terms:** $S_n = n/2(a + l)$ where l = last term

Important Properties:

- In an AP, the middle term is the average of equidistant terms
- If a, b, c are in AP, then $2b = a + c$

4.2 Geometric Progression (GP)

Definition: Constant ratio between consecutive terms.

- First term = a
- Common ratio = r

Important Formulas:

- **nth term:** $a_n = ar^{n-1}$
- **Sum of n terms ($r \neq 1$):** $S_n = a(r^n - 1)/(r - 1)$
- **Sum of n terms ($r = 1$):** $S_n = na$
- **Sum of infinite GP:** $S_\infty = a/(1 - r)$, for $|r| < 1$

Important Properties:

- If a, b, c are in GP, then $b^2 = ac$

4.3 Relation between AP and GP

AP	GP
Terms: $a, a+d, a+2d, \dots$	Terms: $a, ar, ar^2, \dots$
Difference: d (constant)	Ratio: r (constant)
Sum formula uses $n/2$	Sum formula uses $a(r^n - 1)/(r - 1)$

5 Binomial Theorem

5.1 Statement

For any positive integer n ,

$$(a + b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n}b^n$$

In sigma notation:

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

5.2 Binomial Coefficient

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Properties:

- ${}^nC_0 = {}^nC_n = 1$
- ${}^nC_r = {}^nC_{n-r}$
- ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

5.3 General Term

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

Note: T_{r+1} means the $(r+1)$ th term.

5.4 Middle Term

- **If n is even:** $(n/2 + 1)$ th term is the middle term
- **If n is odd:** $(n+1)/2$ and $(n+3)/2$ terms are middle terms

5.5 Special Cases

Expansion	Result
$(1 + x)^n$	$1 + nx + n(n-1)x^2/2! + \dots$
$(1 - x)^n$	$1 - nx + n(n-1)x^2/2! - \dots$
Sum of coefficients	2^n
Sum of even coefficients	2^{n-1}
Sum of odd coefficients	2^{n-1}

6 Determinants

6.1 Definition

A determinant is a scalar value associated with a square matrix.

6.2 Determinant of 2×2 Matrix

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

6.3 Determinant of 3×3 Matrix

Using expansion along first row:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

6.4 Sarrus Rule (Shortcut for 3×3)

1. Write the first two columns again
2. Multiply diagonals down and subtract diagonals up

6.5 Properties of Determinants

Property	Description
Property 1	Value remains unchanged if rows and columns are interchanged
Property 2	If two rows/columns are identical $\rightarrow$ determinant = 0
Property 3	If a row/column is multiplied by k $\rightarrow$ determinant multiplied by k
Property 4	If rows/columns are proportional $\rightarrow$ determinant = 0

6.6 Singular Matrix

A square matrix A is **singular** if $\det(A) = 0$.

7 Matrices

7.1 Definition

A matrix is a rectangular arrangement of numbers in rows and columns.

General representation:

$$A = [a_{ij}]_{m \times n}$$

where $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$

7.2 Types of Matrices

Type	Definition	Example
Row Matrix	Only 1 row	[1 2 3]
Column Matrix	Only 1 column	[1; 2; 3]
Square Matrix	Rows = Columns	[1 2; 3 4]
Diagonal Matrix	Non-diagonal elements = 0	[1 0; 0 2]
Scalar Matrix	Diagonal elements are equal	[2 0; 0 2]
Identity Matrix (I)	Diagonal = 1, rest = 0	[1 0; 0 1]
Zero Matrix	All elements = 0	[0 0; 0 0]

7.3 Matrix Operations

Addition:

- Only possible for same order
- Add corresponding elements

Subtraction:

- Only possible for same order
- Subtract corresponding elements

Multiplication by Scalar:

- Multiply each element by the scalar

Matrix Multiplication:

- Condition: Columns of 1st = Rows of 2nd
- Result: $(m \times n)(n \times p) \rightarrow (m \times p)$

7.4 Special Matrices

7.5 Cayley-Hamilton Theorem

Statement: Every square matrix satisfies its own characteristic equation.

Characteristic Equation: $—A - \lambda I— = 0$

Example: If char. eq. is $\lambda^2 - 4\lambda + 3 = 0$, then $A^2 - 4A + 3I = 0$

Type	Definition
Symmetric	$A = A^T$ ($a_{ij} = a_{ji}$)
Skew-Symmetric	$A = -A^T$ ($a_{ij} = -a_{ji}$, diagonal = 0)
Orthogonal	$A \cdot A^T = I$
Idempotent	$A^2 = A$
Involutory	$A^2 = I$
Nilpotent	$A^k = 0$ for some k

8 Practice Questions (Topic-wise)

8.1 Set Theory

Q1. Which of the following is an empty set?

- (a) $\{x \mid x \text{ is a vowel}\}$
- (b) $\{x \mid x^2 = 4, x \text{ is odd}\}$
- (c) $\{0\}$
- (d) $\{x \mid x + 1 = 1\}$

Answer: (b)

Q2. If $A = \{1,2,3\}$ and $B = \{3,4,5\}$, then $A \cap B$ is:

- (a) $\{1,2,3,4,5\}$
- (b) $\{3\}$
- (c) $\{1,2\}$
- (d) $\{\}$

Answer: (b)

Q3. The number of elements in the power set of $A = \{a, b, c\}$ is:

- (a) 3
- (b) 6
- (c) 8
- (d) 9

Answer: (c) 8

Q4. Write the power set of $A = \{1, 2, 3\}$.

Answer: $P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

8.2 Quadratic Equations

Q5. Solve $x^2 - 7x + 12 = 0$ by factorization.

Answer: $x^2 - 7x + 12 = 0 = (x-3)(x-4) \therefore x = 3, 4$

Q6. Find the discriminant and nature of roots for: $2x^2 - 4x + 3 = 0$.

Answer: $D = b^2 - 4ac = (-4)^2 - 4(2)(3) = 16 - 24 = -8$

$D < 0 \rightarrow$ No real roots (complex roots).

Q7. The sum and product of roots are 5 and 6 respectively. Find the equation.

Answer: $x^2 - (\text{sum})x + \text{product} = 0 \rightarrow x^2 - 5x + 6 = 0$

8.3 Sequence & Series

Q8. The 6th term of an AP is 17 and the 13th term is 38. Find the 19th term.

Answer: $a_6 = a + 5d = 17 \dots(i)$, $a_{13} = a + 12d = 38 \dots(ii)$

Subtracting: $7d = 21 \rightarrow d = 3$, From (i): $a + 15 = 17 \rightarrow a = 2$

$$a_{19} = a + 18d = 2 + 54 = 56$$

Q9. Find the sum of first 12 terms of the series 5, 9, 13, 17....

Answer: $a=5$, $d=4$, $n=12$

$$S_{12} = 12/2[2(5) + (12-1)4] = 6[10 + 44] = 324$$

8.4 Binomial Theorem

Q10. Find the 4th term of $(a + b)^6$.

Answer: $T_4 = T_{3+1} = {}^6C_3 a^{6-3} b^3 = 20a^3b^3$

Q11. Find the coefficient of x^2 in $(1 + x)^5$.

Answer: $T_3 = T_{2+1} = {}^5C_2 (1)^{5-2} (x)^2 = 10x^2$

Coefficient = 10

8.5 Determinants

Q12. Evaluate: $\begin{vmatrix} -2 & 4 \\ 5 & 1 \end{vmatrix}$

Answer: $= 2 \times 1 - 4 \times 5 = 2 - 20 = -18$

Q13. For what value of k will the matrix be singular? $\begin{bmatrix} 2 & 3 \\ 4 & k \end{bmatrix}$

Answer: $\det = 0 \rightarrow 2k - 12 = 0 \rightarrow k = 6$

8.6 Matrices

Q14. If $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, find AB.

Answer: $AB = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 1 \\ -1 \cdot 2 + 3 \cdot 1 & -1 \cdot 1 + 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

9 HPU Exam Important Questions (Unit-I Only)

Based on HPU BCA-0101 Previous Year Papers

9.1 Section-A (5 Marks Each)

Q1. If $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}$, verify that:

$$(A + B)^T = A^T + B^T$$

Q2. Using Binomial theorem, find the value of $(10.1)^3$ up to 3 decimal places.

Q3. Find the solution of equation $\sqrt{1-x} + x = 3$.

Q4. Which term of the sequence 90, 87, 83, ... is zero?

Q5. For what value of k , the line passing through points $(k, 4)$, $(1, 2)$ is parallel to the line passing through points $(3, -1)$, $(2, -2)$?

9.2 Section-E (Compulsory - 1 Mark Each)

Q6. Find the coefficient of x^5 in the expansion of $(x^2 + \frac{1}{x})^7$.

Q7. Construct a matrix of order 2×3 , where $a_{ij} = \frac{i}{j}$.

Q8. What is a singular matrix?

Q9. Find the domain and range of the function: $f(x) = \frac{x+1}{2x+1}$.

Q10. Evaluate the determinant:

$$\begin{vmatrix} 15 & 5 & 20 \\ 1 & 0 & 2 \\ 3 & 1 & 4 \end{vmatrix}$$

9.3 Additional Important Questions

Q11. For what value of k will the matrix be singular?

$$\begin{bmatrix} 2 & 3 \\ 4 & k \end{bmatrix}$$

Q12. Find the eigenvalues and eigenvectors of:

$$\begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$$

Q13. Verify Cayley-Hamilton theorem for:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Q14. Prove that:

$$\det \begin{pmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ca & cb & -c^2 \end{pmatrix} = 4a^2b^2c^2$$

10 Quick Revision Formulas

10.1 Set Theory

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \times B) = n(A) \times n(B)$
- $n(P(A)) = 2^n$

10.2 Quadratic Equations

- Sum of roots = $-b/a$
- Product of roots = c/a
- $D = b^2 - 4ac$
- $x = (-b \pm \sqrt{D})/2a$

10.3 AP

- $a_n = a + (n-1)d$
- $S_n = n/2[2a + (n-1)d]$
- $S_n = n/2(a + l)$

10.4 GP

- $a_n = ar^{n-1}$
- $S_n = a(r^n - 1)/(r - 1)$
- $S_\infty = a/(1 - r), |r| < 1$

10.5 Binomial Theorem

- $(a + b)^n = \sum {}^nC_r a^{n-r} b^r$
- $T_{r+1} = {}^nC_r a^{n-r} b^r$
- Sum of coefficients = 2^n

10.6 Determinants

- For 2×2 : $ad - bc$
- Singular: $|A| = 0$

10.7 Matrices

- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$
- If A is symmetric: $A = A^T$

Exam Tips

1. **Section A & B:** Choose any one question from each
2. **Section C & D:** Choose any one question from each
3. **Section E:** All 10 questions are compulsory
4. **Time Management:** Attempt Section E first (10 marks in 15 minutes)
5. **Formula Sheet:** Write all formulas at the top of your answer sheet

All the Best for Your HPU BCA Semester-I Mathematics Examination!

— End of Notes —